

Use this document to detail specifications/requirements for an item and/or ancillary services. ⓘ

TITLE: Replacement high-performance cesium beam tube for 5071A frequency standard ⓘ

Requesting Lab/Division/Group: PML/688.40/Time Realization and Distribution

General Statement of Need ⓘ

NIST operates several 5071A cesium beam atomic frequency standards as part of the ensemble atomic time scale that generates the UTC(NIST) time and frequency standard. The 5071A's cesium beam tube has a finite lifetime, but can be replaced without purchasing an entirely new atomic clock chassis. At this time, NIST requires one cesium beam tube to replace a tube which reached its end-of-life condition. NIST will perform installation of the new cesium beam tube. Note: "high-performance" here is technical language and refers to two variants of cesium beam-tube design. The "long-life" tube variant consumes cesium at a slower rate, at the cost of roughly a factor of 5 in timekeeping performance compared to the "high-performance" variant. The equivalent performance metrics are listed below as specifications.

Items and/or Ancillary Services ⓘ

Item 1: Replacement high-performance 5071A cesium beam tube

Quantity: 1

Specifications: Specification 1.1. Cesium beam tube must be electrically and mechanically compatible with the 5071A high-performance cesium beam frequency standard.

Specification 1.2. Cesium beam tube, when operated in a 5071A high-performance cesium beam frequency standard, must meet the frequency stability specifications of $\leq 5 \times 10^{-12}$ at 1 second averaging intervals and $\leq 1 \times 10^{-14}$ at 30 day averaging intervals.

Specification 1.3. The cesium beam tube must be new.

Separately Priced Options and Services:

Item 2: Optional: recycling service for old cesium beam tube

Quantity: 1

Specifications: Specification 2.1. If not included in Item 1, then separate priced recycling service fee for returning old cesium beam tube to manufacturer for safe disposal and recycling.

[List any other item(s) or service requirements here using the same format above.]

Service Requirements / Item Specifications

Section ⓘ	Requirements/Specifications ⓘ
Planning	☐ List any visits required to allow potential vendors to see site specific constraints.
Considerations	☐ [Explain requirement here]
☒ N/A	☐ US citizen required on-site (foreign nationals require preregistration 30 days prior)
	☐ Vendor rep on-site more than three days

Requirements / Specifications: Replacement high-performance cesium beam tube for 5071A frequency standard 1

☐ Other considerations: [List other considerations to ensure an accurate quote]

Shipping & Delivery

☐ N/A

[Enter shipping and delivery information by item here.]

Shipping criteria:

☐ Partial delivery acceptable

☐ Direct delivery to building required

[Explain need for requirement here]

Building has: ☐ Loading dock

☐ Containerization preference: [Choose container]

Other requirements: [List any other requirements here]

Delivery date criteria:

☒ Delivery of goods, installation, and training shall be completed no later than: 24 weeks ARO

☐ Deliver no earlier than: [Insert date here]

☐ Other: [List other shipping, delivery, or specil requirements]

Electronic Media

☒ N/A

[Enter electronic media requirements by item here.]

☐ Software

☐ Electronic manuals

☐ License Key

☐ Other: [List any other Electronic Media requirements here]

[Enter email address for software delivery here]

Installation

☒ N/A

[Enter installation requirements by item here.]

☐ Vendor technician on site

☐ Rigging

☐ Uncrating / Unpackaging

☐ Removal of packaging material

☐ Equipment set up

☐ Start-up services

☐ Turn key installation

☐ Other [Enter details here.]

Note: When installation is required, TPOC or COR shall be the coordination point.

Facility / Utility Considerations

☒ N/A

[List site specific utility considerations by item here.]

☐ OFPM Work Order Number(s)

[List any Work Order Number(s) here]

☐ Power available

Voltage: [Enter voltage here] (V)

Max Current: [Enter max current draw here] (A)

Phase: [Choose phase]

Frequency: [Typically 60Hz] (Hz)

NEMA Plug Type: [Enter plug type here] (i.e.: NEMA L5-20P)

Environment: [Choose environment]

Other power consideration: [List any other power considerations here]

☐ Utility connections: [List relevant available utilities, pressure(s), and flowrate(s)]

☐ Other considerations: [List other considerations to ensure an accurate

quote]

Warranty ⓘ

☐ N/A

Base Manufacturer Warranty is acceptable

[If you require an Extended Manufacturer Warranty, describe the warranty requirements per item here.]

What is included?

☐ Labor ☐ Parts ☐ Travel ☐ Cost and liability for returns

☐ Other: [Enter other requirements here]

☐ Warranty length: [Enter warranty length here]

Period Length:[Ex. month, 6 month, year etc.]

Training

☒ N/A

[Training requirements (who, when, what, where) here. If on-site, is travel necessary?]

☐ On-site

☐ Off-site location: [Enter location(s) here]

☐ Number of trainees: [Enter number of trainees here]

Scope of training:

☐ Operations ☐ Maintenance ☐ Troubleshooting ☐ Safety

☐ Other: [Enter other scope requirements here]

☐ Training materials to be provided: [Describe requirement here]

☐ Other: [Enter other training requirements here]

Note: When training is required, TPOC or COR shall be the coordination point.

Inspection and

Acceptance

☐ N/A

[Enter inspection requirements per item here]

☒ Planned government acceptance testing (7 business days)

☐ Gov't acceptance expected to take longer than 7 business days: [Duration]

☐ Inspection and acceptance plan: [Explain here]

☐ Other: [Enter other requirements here]

Maintenance ⓘ

☒ N/A

[Describe your maintenance requirements per item here]

Period Length: [Ex. month, 6 month, year etc.]

What is included?

☐ Labor ☐ Spare parts ☐ Consumables ☐ Travel

☐ Cost and liability for returns ☐ Software updates ☐ Manuals

☐ Service plan length: [Enter service plan length here]

☐ Other: [Enter other requirements here]

☐ Requirements: [Enter requirements here] (i.e.: onsite within 72 hours, etc.)

☐ Option Period 1 ☐ Option Period 2

☐ Option Period 3 ☐ Option Period 4

Government

Furnished

Material /

Equipment

[List any government furnished material/equipment per item here]

☐ Samples for testing

☐ Government equipment which will leave gov't site (TPOC must coordinate with division property custodian)

Serial Number: [Enter Serial Number here]

☒ N/A

Description: [Enter description here]

NIST property number: [Enter NIST property number here]

☐ Other: [Enter other details here]

Travel

[Describe any travel requirements per item here]

☒ N/A

☐ Explain: [Explain here]

☐ Other: [Enter other requirements here]

Vendor: To the maximum extent possible, include as part of appropriate sections above (maintenance, repair, etc.). Otherwise, it will be a separate line item, reimbursable per the Federal Travel Regulation.

Provide suggestions to improve this form by clicking [here](#).